Disclosure Notes Relation to Financial Quality Metrics in Going-Concern Companies: Evidence of Non-Association

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Abstract

Does a company's writing style and form indicate problems in its financial statements? My original study broadly examined this potential association between several disclosure note types and financial quality signals and found a null. I decided to dive deeper into these disclosure notes and focus specifically on going concern companies, as these distressed companies would most likely indicate any correlation between writing style and indications of financial issues. I built a research instrument that aggregated 161,469 footnotes from 3,253 public companies to aid in my study and analysis of this question. This tool measures the disclosures' distinctiveness compared to other companies and the complexity of their language. These measures are then compared to the Dechow F-Score and Beneish M-Score to assess their relation to a company's financial condition and potential manipulation risk. The study found no association between the way companies write their financial disclosures and their financial condition. To ensure the null wasn't something I overlooked, I conducted a systematic screening of 460 unique tests across all subgroups and measures and applied FDR and Bonferroni statistical corrections for the extensive number of tests. The 3 associations that survived ($\rho \leq 0.16$) were essentially meaningless. Overall, this study suggests even companies in distress don't reflect their financial conditions in their disclosure notes. This research instrument was built using Claude Code and reflects AI's ability to enhance and accelerate the research process by enabling data aggregation, testing, and review at scale, while emphasizing the importance of proper safeguards when using AI to prevent false discoveries and maintain data integrity.

1. Introduction

Going concern warnings are issued when the company's auditors believe there is substantial doubt that the company will continue for the next 12 months. This is one of the most serious statements an auditor can give to a company. This study focuses on companies that have been issued a going concern warning due to the severity of this warning and their financial condition issues; if there is any disclosure language that reflects financial issues, these public companies are where it should be the most visible.

This study is an extension of my original null regarding whether disclosure readability and distinctiveness are associated with a company's financial condition. My original null prompted me to examine disclosures more deeply and focus specifically on companies that have been issued a going concern warning. My intuition is that going concern companies that are essentially in financial distress would be the most relevant and testable for this study. From my full corpus this sample is comprised of 4,186 company-years of going concern firms. I created Disclosure Atlas to aid in this study. Disclosure Atlas is a browser-based semantic search research instrument that maps out 161,469 disclosure footnotes from 3,253 public companies. This tool enabled me to conduct language and financial quality measures at scale. Language was analyzed for readability and distinctiveness through two measures. Readability was calculated for the disclosure notes through the Gunning Fog score. Distinctiveness was assessed by embedding each footnote, computing the "average," and measuring the cosine distance of each footnote from the average; this is the cosine distance measure in Disclosure Atlas. The financial quality measures used were the Dechow F-Score and Beneish M-Score. The Dechow F-Score measures a company's probability of materially misstating its financial statements, while the Beneish M-Score measures potential earnings manipulation through 8 financial ratios.

Through this study, I found that companies that had been issued a going concern warning did not have meaningful significant associations when assessing the connection between their disclosure language and their financial health. For example, of the 460 tests that ran across a systematic screen, only 3 minuscule associations survived, none of which involved the financial scores.

My study makes 3 contributions. Firstly, even among distressed firms, indicated through being issued a going concern warning, these firms' writing styles and financial metrics are not interlinked. Secondly, this non-association holds up under systematic screening and has undergone several reviews and verifications. Finally, I introduced safeguards to protect against cheating and selective reporting. This is incorporated by implementing required corrections, mandatory reporting of everything, and a hash record that timestamps the test and locks the test and metrics it is testing before the results are displayed.

2. Background

This study combines two areas of accounting research that are not typically combined: analysis of financial disclosure language and quality metrics.

The text analysis of disclosures was focused on how readability and distinctiveness of these disclosures correlate to the company's financial metrics. Readability and complexity were measured through the Gunning Fog Index, which is a metric that quantifies the estimated level of education needed to understand the text (Li 2008; see Loughran and McDonald 2016 for a survey of textual analysis in accounting and finance). For example, the average Wall Street Journal article has a score of 10-12 while a public company's 10-K usually scores between 18-20. Distinctiveness assesses how uncommon the company's disclosure language is relative to

similar companies. This is measured by turning each footnote into an embedding to quantify it relative to other footnotes. Then the average footnote was found by taking the average of all the embeddings and measuring how far each footnote is from the "average" using the cosine distance (Hanley and Hoberg 2010; Brown and Tucker 2011). The larger the distance, the more irregular the company's disclosures are, relative to similar companies.

The financial quality analysis is measured through two models. The Dechow F-Score (Dechow, Ge, Larson, and Sloan 2011) and Beneish M-Score (Beneish 1999). The Dechow F-Score assesses the probability of a material misstatement by analyzing financing activities, performance, and accrual quality. The Dechow F-Score has a baseline score of 1.0, which represents the risk of a normal company, while higher values mean an above-average risk. The Beneish M-Score uses eight ratios to gauge the probability that companies are manipulating their earnings. These eight ratios are sales growth, change in receivables, asset quality, gross margin, depreciation, leverage, expenses, and accruals. A score of above -1.78 indicates the financials share common characteristics of known manipulators. Both of these models are widely used but come with well-documented limitations. Some limitations are sensitivity to large business events like a large divestiture or acquisition and high false positive rates. These models are screens and are not verdicts; thus, this study treats them as such.

This study is an extension of my original null which focused on the association between several disclosure note types and financial quality signals. This was examined by comparing the relationship between companies that had been enforced by the SEC and comparing them to clean firms. My finding was that there was no meaningful difference in disclosure language between the two. This prompted the question of whether disclosure language tracks the financial condition of companies at all.

3. Data

The total corpus studies 161,469 disclosure footnotes that were drawn and ingested from the SEC EDGAR filing system. These disclosure footnotes are comprised of 3,253 U.S. public companies across 6 disclosure types. Disclosure types were composed of going concern, related-party, revenue recognition, risk factors, management's discussion and analysis, and critical audit matters. The corpus' timeline ranges from 1996-2026, but ~97% of the corpus is from 2016-2026. This is due to data being drawn from the most recent 10 years of annual filings for each company. The range from 1996-2016 is due to some companies' most recent filings being older.

I identified going concern footnotes using a rule-based extractor that's comprised of two methods. The first and simpler method utilizes inline XBRL tags already labeled from the SEC. But only roughly 18% of footnotes had XBRL tags. The other method was a keyword heuristic; the rule for this method in order to extract the going concern footnote was to require the phrase "going concern" and have words that indicate the auditor had substantial doubt that the company would survive the next 12 months. I then reviewed a manual sample of the extractions and found

that ~70-80% of the footnotes were correctly identified. The ~20-30% false positives were due to three reasons. Firstly, standard descriptions that explain what a going concern is; second, hypothetical risk language used to discuss if something happened, and the potential issues it would incur. Finally, certain XBRL tags would get split (~3.6%), and some split versions of the tag would not discuss actual going concern content. In order to combat this issue, I also built a stricter and higher-precision subset that imposed stricter rules. These stricter rules include requiring substantial doubt language and excluding the potential issues of the standard descriptions and hypothetical risk language reasons. The higher-precision subset yielded 2,572 company-years and 896 companies. Overall, I reached the same conclusion with both the stricter subset and the original cohort.

To establish my unit of observation, I aggregated at the company-year level. This is due to companies mentioning going concern several times in a single filing; this helps average the language measures and prevents a company from being counted several times in a given year. This resulted in 4,291 company-years spanning 1,408 companies from the time range of 1998-2026. Two restrictions to note: I excluded 15 company-years in the data validation phase of the study due to a merging issue. Also, I restricted the study to 2010-2026, which dropped 90 company-years from the small data trail before 2010. The final cohort was 4,186 company-years and 1,373 companies. Of the 4,186 company-years, only a subset of these can compute the Dechow F-Score (1,323 company-years, ~31.6%) and the Beneish M-Score (1,120 company-years, ~26.8%) financial scores. This is due to the criteria these scores require to be computed. These going concern companies frequently have gaps in their reporting and overall incomplete financial statements, which prevents being able to use these screening models, highlighting the limitations of these models. The most distressed companies are the ones that these standard screens can't even compute.

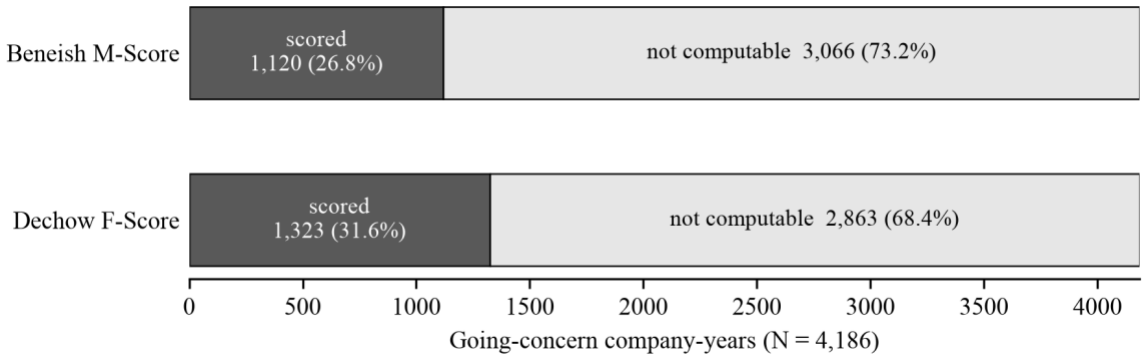


Figure 1. Financial score coverage of the going-concern cohort (N = 4,186 company-years). For the Beneish M-Score, only 1,120 company-years (26.8%) and 1,323 company-years (31.6%) for the Dechow F-Score had enough data to compute their respective scores.

4. Methodology

4.1 Measures

The disclosures are measured for their level of readability and distinctiveness in various ways. Readability is measured through the Gunning Fog Index. This is a readability metric that estimates the amount of formal education a reader needs to comprehend the text on the first reading. This helps compare the complexity of the disclosure. The Gunning Fog Index is the most used readability measure in this area, but it is debated for financial text. Loughran and McDonald (2014) show that Fog does not work well for business disclosures because it counts common multisyllabic business words like corporation and operations as complex. I still use Fog because it is the standard benchmark and keeps my results comparable to other studies, but I treat it as a rough measure rather than an exact one. Distinctiveness measures how unique a disclosure's language is from similar companies. This was calculated by embedding each footnote with bge-small-en-v 1.5, a pretrained embedding model (384 dimensions, L2-normalized), essentially, each footnote was converted into 384 numbers to quantify its relation to other footnotes. Each embedding was normalized so that comparison only measures similarity in meaning and not the length of the text. Then for each peer group, I computed the average embedding. The cosine distance is each footnote's embedding to the peer group's average embedding; this reflects the distinctiveness of each footnote. The financial quality metrics were the Dechow F-Score using its seven inputs and the Beneish M-Score, which utilizes eight ratios. These metrics were computed from SEC XBRL financials.

4.2 Unit of Observation and Aggregation

This study was conducted on a company-year level. This was to account for when a company would have multiple instances of going concern language in a single year. Collapsing to one record per company-year helped prevent overweighting certain company years that mention going concern multiple times in that given year.

4.3 Statistical Approach

A well-known characteristic of the Dechow and Beneish models is that near-zero denominators in their component ratios cause extreme outliers. This is a characteristic of the formulas themselves and not due to the data. This required me to change the statistical approach. Due to this outlier issue, I could not use standard deviations, means, and Pearson correlations, as this would skew the results and would not give an accurate representation of the sample. But rather use IQR, medians, Spearman rank correlations for association, and Wilcoxon rank sum over a t-test. Utilizing rank-based methods allows insight into the data without being impacted by extreme outliers due to relying on the order of values.

4.4 Systematic Screening

I developed a systematic screen to find any associations that show up in specific subgroups, rather than only using one test. The screen works by pairing every financial and disclosure measure across all subgroups. These subgroups are composed of industry, entire cohort, enforcement status, and 5-year buckets. The tests were enumerated through a deterministic rule, and there were 460 total tests. Before the screening takes place, I implemented several safeguards and corrections. For example, every test in the family is reported; this is so any reader can see the full report and total tests run, and the number of tests that failed. Also, each screening is recorded with a hash that marks what is being tested and when the test was conducted. This is to prevent secretly adding, removing, or reframing tests that didn't work after seeing the outcome. Every result reports its effect size with its p-value because with a large enough sample size, an association can be statistically significant while practically being meaningless (Cohen 1988). I incorporated two mandatory corrections across all tests: the Benjamini-Hochberg FDR and the Bonferroni. FDR controls the expected proportion of false positives among results that appear to be significant. The Bonferroni is the strictest by dividing your p-value cutoff by the total number of tests. Any survivors require independent review and confirmation. These safeguards and corrections are intended to make selective reporting deliberate rather than a default and help prevent against p-hacking.

5. Results

5.1 Descriptive Statistics

The entire cohort is summarized in Table 1. Table 1 shows that the median Gunning Fog Index is 19.6 with an IQR of 4.7. This reflects that these disclosures require about a graduate-level reading level in order to understand them. The median distinctiveness is 0.115 with an IQR of 0.060. The small median and narrow IQR display that these disclosures are fairly similar to their peers and reinforce how they are largely standardized from one another. For financial quality scores, the median for the Dechow F-Score is 0.57, IQR 0.88 and the Beneish M-Score is -2.74, IQR of 1.99. Both medians are below their thresholds. Because both medians are below their thresholds (Dechow F-Score baseline of 1.0 and Beneish M-Score of -1.78), the typical going-concern company is not flagged as a likely manipulator by either model.

Table 1. Descriptive statistics, going-concern cohort (company-year level)

Measure	N	Median	IQR
Gunning Fog Index (readability)	4,186	19.6	4.7
Distinctiveness	4,186	0.115	0.060
Dechow F-Score	1,323	0.57	0.88
Beneish M-Score	1,120	−2.74	1.99

Note. Due to the extreme outliers of the Dechow F-Score and Beneish M-Score, median and IQR are used in place of mean and standard deviation. The financial scores are only available for a subset of company-years (31.6% Dechow, 26.8% Beneish). Thresholds for each model: Dechow F-Score 1.0, Beneish M-Score -1.78

5.2 Disclosure Form and Financial-Quality Signals Are Not Associated

The Spearman correlation of the four measures is represented in Table 2. As shown, every association is negligible. Readability with Dechow ($\rho = 0.03$), with Beneish ($\rho = 0.02$). Distinctiveness with Dechow ($\rho = -0.03$), with Beneish ($\rho = 0.06$). There were two trivially small significant associations at the five percent level. The Dechow with Beneish ($\rho = 0.10$, $p = 0.002$) and distinctiveness with Beneish ($\rho = 0.06$, $p = 0.045$). Neither survived the mandatory correction measures. When looking at the association between disclosure form and financial quality, they are not meaningfully correlated in any way. Figure 2 represents this correlation graphically.

Table 2. Spearman rank correlations among principal measures (N: Fog - Dst. 4,186, Fog – Dechow 1,323, Fog - Beneish 1,120, and Dechow – Beneish 871).

	Fog	**Distinct.**	**Beneish M**	**Dechow F**
Gunning Fog	1.00	0.02	0.02	0.03
Distinctiveness	0.02	1.00	0.06	−0.03
Beneish M-Score	0.02	0.06	1.00	0.10
Dechow F-Score	0.03	−0.03	0.10	1.00

Note. Two nominally significant associations between Dechow – Beneish ($\rho = 0.10$, $p = 0.002$) and Dst. – Beneish ($\rho = 0.06$, $p = 0.045$). Both of which do not survive the FDR correction.

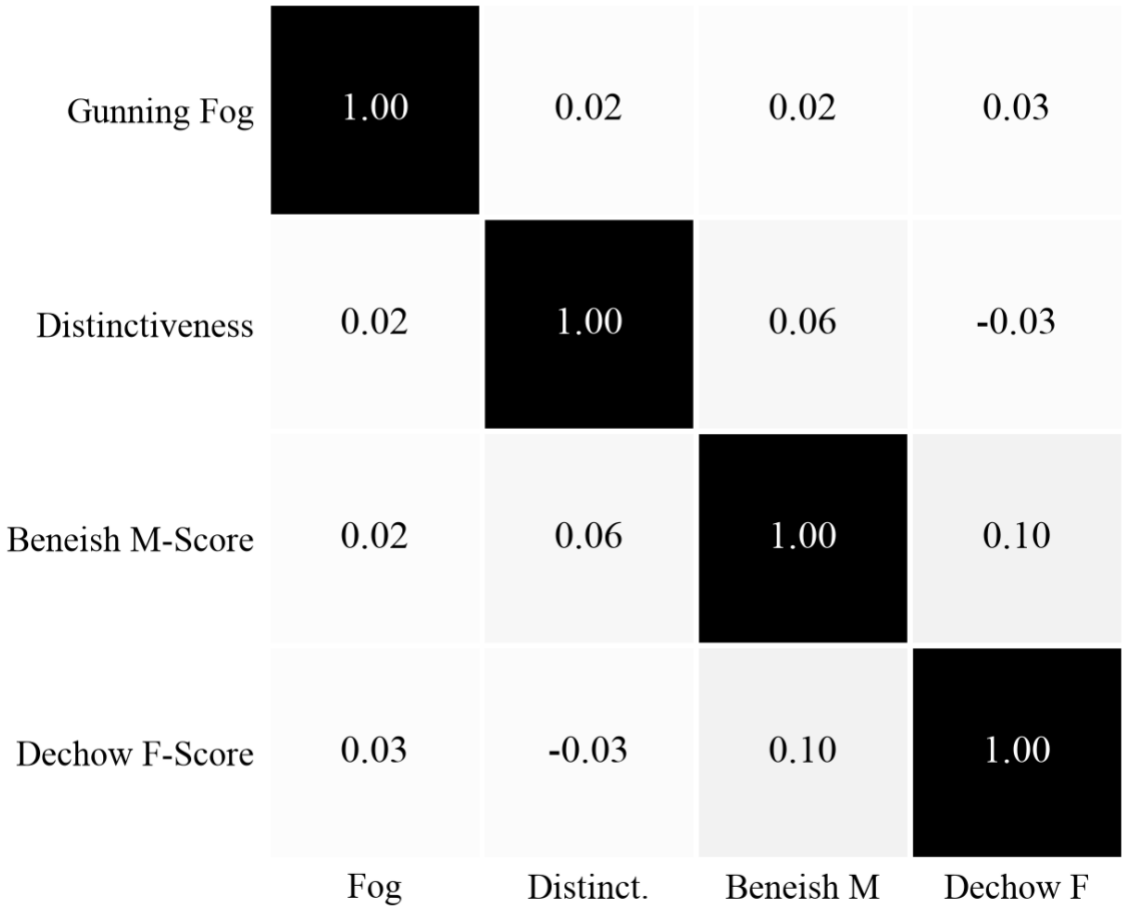


Figure 2. Correlation matrix visualization

Spearman rank correlations among principal measures shown graphically, all shading is proportional to |ρ|.

5.3 Enforced and Clean Firms Are Indistinguishable

Going concern companies with a history of SEC enforcement are compared to those not enforced in Table 3. I examined whether enforced and non-enforced firms write differently and if their financial scores differ. Through testing, I discovered they are practically indistinguishable. When assessing readability through the Gunning Fog Index there is a nominally significant difference at the company-year level (19.6 vs. 19.4, $p = 0.026$), but the size of the difference is negligible. When this is examined over companies rather than company-years, the result is not significant ($p = 0.11$). No other measures have differences that are significant. Please note these results are suggestive rather than conclusive due to the enforced subsample only being 179 company-years and 57 firms. This causes firm count to influence results rather than depending on company-year.

Table 3. Enforced versus non-enforced going-concern firms (medians)

Group	N (CY / firms)	Fog	Distinct.	Beneish M	Dechow F
Not enforced	4,007 / 1,316	19.6	0.115	−2.76	0.57

Enforced	179 / 57	19.4	0.112	−2.64	0.55

Note. Results from Wilcoxon rank-sum tests, two-sided: Gunning Fog Index p = 0.026 (nominally significant), distinctiveness p = 0.674, Dechow p = 0.943, Beneish p = 0.106. Enforced subsample is comprised of 179 years in 57 companies. (58 scored company-years from 26 companies for Beneish)

5.4 Non-Association Is Robust Across a Systematic Screen

I developed a systematic screening tool to examine all subgroups and measures to determine if a specific subgroup had any disclosure and financial associations. The results of this systematic screen are displayed in Table 4 and were comprised of 460 tests. After the 460 tests were run, the two mandatory corrections were applied. The results were that three survived the FDR correction while only one survived the Bonferroni, the largest survivors size was $\rho = 0.155$. From the FDR correction, the three survivors all show one association between distinctiveness and one of the eight Beneish ratios (Total accruals over total assets). This appears in nested subgroups and the overall association is weak. When compared to the full corpus, it has the opposite direction, which indicates it is population-specific. Results found no surviving association involves the composite Dechow and Beneish screens and nothing displays meaningful significance. Overall, the systematic screens reinforce that there is no strong association between disclosures and financials in the pre-committed family.

Table 4. Systematic screen summary (going-concern cohort)

Quantity	Value
Candidate tests enumerated (34 pairs × 258 subgroups)	8,772
Tests in family (pairwise $N \geq 30$, non-constant)	460
Excluded by pre-specified rule (listed, not dropped)	8,312
Survive Benjamini-Hochberg FDR ($q < 0.05$)	3
Also survive Bonferroni	1
Largest survivor effect size	$\rho = 0.155$

Note. Every test is at the company-year level using a two-sided Spearman correlation. All 460 tests have the FDR and Bonferroni corrections applied. The 3 surviving tests from the FDR correction collectively indicate the association between distinctiveness and TATA ratio from the Beneish calculation. This only appears in nested subgroups and the direction is flipped when examining the full corpus.

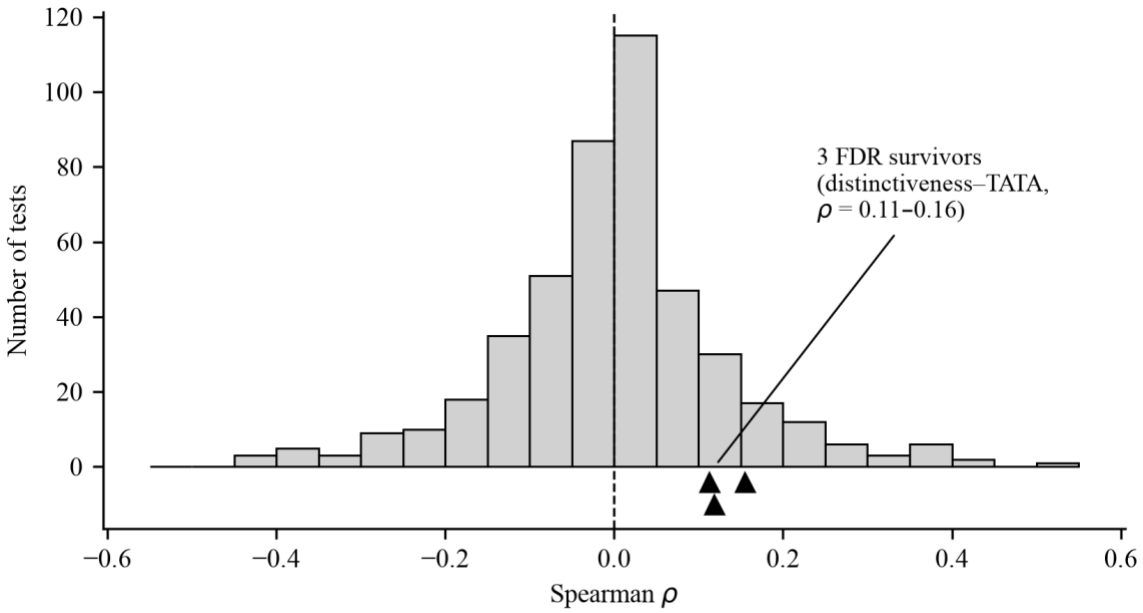


Figure 3. Distribution of Spearman ρ across all 460 tests

This displays the spread of the Spearman ρ values across all 460 tests in the systematic screening (Table 4). The 3 FDR survivors are indicated via triangles. All of which reflect the same distinctiveness-TATA association ρ = 0.11–0.16). The dashed line marks ρ = 0. All extreme values are from small subgroups and did not survive the correction.

5.5 Firm-Level Robustness

Testing was conducted from the 4,186 company-years from 1,373 companies. Companies typically write similarly and have low volatility in their financial ratios year over year. This results in company-years from the same company not being genuinely independent data points. To check and amend this, I redid the analysis at the company level. Each company was collapsed to a specific data point by taking the median of its values across all of its company-years. This stricter test resulted in a stronger non-association. All four disclosure and language correlations were reduced to essentially zero. Readability with Dechow ($\rho = 0.01$, $p = 0.83$), with Beneish ($\rho = 0.02$, $p = 0.62$). Distinctiveness with Dechow ($\rho = 0.002$, $p = 0.97$), with Beneish ($\rho = 0.05$, $p = 0.26$). This stricter test shows that the nominally significant distinctiveness and Beneish association from before was a result of non-independence, not an actual relationship due to disappearing at the company level. The Dechow and Beneish association ($\rho = 0.12$, $p = 0.015$), this proves the stricter test still reflects genuine relationships. Also, I ran a company-clustered bootstrap with 95% confidence intervals (4,000 resamples).

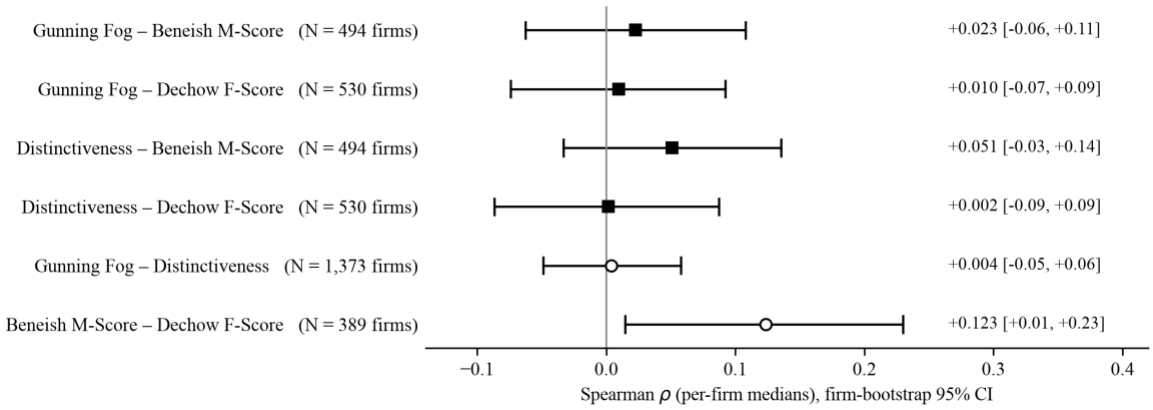


Figure 4. Firm-level Spearman correlations with 95% confidence intervals

Displays Spearman correlations on company level with their 95% confidence intervals, these were discovered using the bootstrap with 4,000 resamples on per company medians (see Section 5.5). The filled in black squares are the disclosure and financial pairs, The circles are within family pairs. As you can see the four disclosure and financial pairs cross zero, meanwhile the Dechow and Beneish interval doesn't.

6. Limitations

Several limitations bound the interpretation of these results.

First, the disclosure-form measures are limited in scope. The two measures used to analyze disclosures were readability (Gunning Fog Index) and distinctiveness; both of these encapsulate how the disclosure reads and how unusual it is. They do not measure the meaning of the disclosure. Finding no association for these two structural features to financial signals does not imply the meaning of disclosures is unrelated to financial condition. Deeper measures like topic models and sentiment could potentially reveal associations these measures cannot, and are the next natural steps. This concern is stronger because the Gunning Fog Index itself is debated for financial text (Loughran and McDonald 2014). If Fog measures readability with error, that error would push the correlations toward zero. This means part of my null could come from measurement limits rather than a true non-association.

Second, the financial-quality measures are imperfect proxies. Both financial screening tools the Dechow and Beneish, have well-documented false-positive rates and are not direct measurements of a company's financial condition. My results from this study are best interpreted as no association between my disclosure language measures and two well-established financial quality screens.

Third, examination of disclosures, not outcomes. The cohort analyzes firms issued with a going-concern warning but does not track whether the doubt came true. It is not apparent if these firms got delisted, acquired, failed, etc. Linking these disclosures would require additional external data, and reflects a natural extension.

Fourth, the going-concern label is imperfect, and measurement error may attenuate observed associations. The extraction tool in Disclosure Atlas was roughly 70-80% accurate at

the footnote level for the full cohort. Potential label noise could push the correlations towards zero, it's technically possible that a real association was masked due to contamination. However, in the stricter test the non-association still holds (higher-precision subset, see appendix), this reduces my concern for this limitation.

Fifth, Unavailable financial-quality scores for a majority of company-years. Due to incomplete information from many of these distressed companies, the financial quality scores could not be computed for the full cohort. Financial quality scores were calculated 31.6% for Dechow and 26.8% for Beneish. This limitation displays how my sample is a subset of the full cohort, demonstrating how these models to assess financial condition are not even applicable to certain distressed firms.

Sixth, the enforcement comparison rests on a small number of enforced firms. Enforcement firms comprise 57 companies and 179 company-years, which is overall lacking statistical power. This is offered as suggestive rather than definitive. (See section 5.3)

Seventh, the analysis is conducted using a single instrument. Disclosure Atlas was developed and used for cohort counts, financial scores, systematic screening, and the correlation statistics. All key numbers were recalculated and verified independently. When rerunning a supervised probe, there was a weak ability to tell apart clean and enforced companies (AUC 0.64-0.69). This most likely reflects residual time period and industry variances.

7. Discussion

The overall finding is that the going concern companies disclosure form reflects non association with its relationship to financial quality signals. Neither readability nor distinctiveness can meaningfully distinguish clean from enforced companies. These findings hold at the company-year level and in the 460-test systematic screening. The few survivors reflect weak population-specific associations with accrual components and reversals in the full corpus. Overall, disclosure form and financial quality signals largely capture independent information.

This impacts the common assumption of how the way a company writes its disclosures encodes information about the underlying financial condition. My findings suggest that financial disclosure form is so standardized that it doesn't reflect a company's specific financial reality, but rather displays companies' general professional conventions of disclosure norms and the legal templates that they follow. This is further supported by distinctiveness' narrow IQR of 0.060. The IQR points to disclosures clustering to their similar peers with little to no divergence from the norm. These claims are bound by the focus on structural language properties (readability and distinctiveness) rather than semantic content. Also this study assesses the association between the financial models used (Dechow and Beneish) rather than the direct financial condition of the company. A worthwhile next step would be using content-based measures to see if I reach the same conclusion of non-association.

In addition to these specific findings, this project displays a methodological shift with broad potential implications for accounting research. Disclosure Atlas was built and used to construct the corpus, measures, and the screening framework. Through Disclosure Atlas, the screening framework was able to run all 460 tests in a matter of seconds. All of which was built by myself with Claude Code. Historically, researchers have spent a vast amount of time gathering, cleaning, and preparing data. Using AI substantially lowers the data wrangling phase of research. With that additional capacity, researchers can spend more time using their judgment and exploring more research questions. AI can be used to develop custom research tools and run analyses at scale much faster. If accounting researchers provide public access to their custom research tools, this would help improve the reproducibility of studies, further propelling the researchers' field forward. However, this ability for speed and scale increases the importance of safeguards and methodological discipline. The ability to run many tests easily increases the rate of false discovery through selective reporting. Due to this, I implemented several safeguards and preventions into Disclosure Atlas. For example, mandatory corrections, entire family reporting, and advanced commitment on a test with a hash ID. These safeguards become crucial rather than optional. I believe the value in using AI in research isn't to replace the need for human judgement but rather to directly incorporate it into the researcher's workflow. For example, AI can aid in developing tools, data wrangling, performing testing, and contributing to better insights. Human judgement and oversight remain essential because research requires interpretation and proposing hypotheses, both of which require the researcher to exercise judgement on what is meaningful and significant.

8. Conclusion

The way companies issued with a going concern warning write their disclosures and its connection to financial quality metrics appears to be largely unrelated. This non-association is displayed across all 460 tests from the systematic screening and when analyzing clean and enforced companies. The evidence from these results point to disclosure form (readability and distinctiveness) displaying the typical disclosure note conventions companies follow more than their financial condition. These discoveries were found using a research instrument I was able to build using Claude Code, demonstrating the vast possibilities in how AI can change accounting research as long as proper safeguards are established.

AI-Use Disclosure

The research instrument developed and used for this study was built with Claude Code (tool, pipeline, database, systematic screener, and analysis). I planned and directed research design, cohort definitions, measure choices, and interpreted the results. Due to AI's risk of hiding errors I verified and computed the values separately and checked with third party libraries. Financial scores were recomputed from the raw inputs with zero mismatches (11,717 Dechow, 31,391 Beneish company-years). Correlation and screening stats were rederived against independent

scipy routines; the previous embedding was also reproduced with separate code. I also independently analyzed and verified results utilizing R. These verifications are not independent of me. The multiple comparison safeguards measures are there due to AI being able to run many tests; I believe implementing this helps against false discovery and required them to be in the tool. No findings were written without me verifying and checking the numbers, any errors are my responsibility.

Appendix: Robustness

Higher-precision label subset. The analysis on the stricter subset of going concern company-years (disclosures excluding the accounting standard descriptions and conditional risk factor phrasing and containing substantial doubt language) was comprised of 896 companies with 2,572 company-years. The same finding of the non-association occurs in this subset. Readability with Dechow ($\rho = 0.04$, $p = 0.30$), with Beneish ($\rho = -0.02$, $p = 0.65$). Distinctiveness with Dechow ($\rho = -0.03$, $p = 0.36$), with Beneish ($\rho = 0.04$, $p = 0.32$). The stricter subset displays that the association between Dechow and Beneish got weaker, which reflects that the weak agreement fades with distressed companies.

Screen survivors. The 3 FDR from the systematic screening reflect the same association between distinctiveness and total accruals (full-cohort $\rho = +0.12$, firm-clustered 95% CI [+0.05, +0.19]. The higher precision subset $\rho = +0.10$, 95% CI [+0.004, +0.203]. In the full corpus this pair is in the opposite direction. This implies the association is specific to the population. Overall, both sizes are below meaningful significance.

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Data Availability

Disclosure Atlas is a browser application available at disclosure-atlas.vercel.app with a project page at simplydose.app/disclosure-atlas. The source code, measures, systematic screen specification, independent evaluation harness, and replication code that reproduces every table and figure in this paper are available at github.com/SimplyDose/disclosure-atlas under an MIT license (disclosures are from public SEC EDGAR filings). Correspondence: Joshua Munger, University of Missouri, School of Accountancy.